

MYSQL DBA CONFIGURATION

1. High CPU — what can cause it ?

Common reasons:

- Expensive/long-running SQL queries
- Missing indexes → full table scans
- Bad execution plans
- Large joins/sorts
- Too many concurrent queries
- High connection/thread activity
- Heavy INSERT/UPDATE/DELETE
- Replication SQL thread doing heavy work
- Backup/compression running
- MySQL configuration causing excessive work

What I check

First Linux:

top
or:
htop

Find the process consuming CPU:

```
ps -eo pid,user,%cpu,%mem,cmd --sort=-%cpu | head -20
```

If mysqld is high, go into MySQL:

```
SHOW FULL PROCESSLIST;
```

Look for:

Time
State
Info

Then investigate expensive queries using Performance Schema / sys schema.

For a suspicious query:

```
EXPLAIN <query>;
```

Example

CPU 95%

↓

mysqld = 90%

↓

Find active queries

↓

UPDATE/SELECT running for long time

↓

```
EXPLAIN
```

↓

Full table scan

↓

Missing index

↓

Fix index/query

Don't assume high CPU means "add CPU." First find what MySQL is doing.

2. High Memory — what can cause it ?

Common reasons:

- InnoDB buffer pool too large
- Too many connections
- Large per-connection buffers
- Large sorts/joins
- Temporary tables
- Memory-heavy queries
- Performance Schema memory
- Replication-related memory
- Other processes on the server
- Memory leak in another application/process

What I check

Linux:

```
free -h
top
ps -eo pid,user,%mem,rss,vsz,cmd --sort=-%mem | head -20
```

Check swap:

```
free -h
swapon --show
vmstat 1
```

Then MySQL:

```
SHOW VARIABLES LIKE 'innodb_buffer_pool_size';
```

Connections:

```
SHOW GLOBAL STATUS LIKE 'Threads_connected';
```

Maximum:

```
SHOW VARIABLES LIKE 'max_connections';
```

Temporary tables:

```
SHOW GLOBAL STATUS LIKE 'Created_tmp%';
```

And on MySQL 8:

```
SELECT
    event_name,
    current_number_of_bytes_used
FROM performance_schema.memory_summary_global_by_event_name
ORDER BY current_number_of_bytes_used DESC
LIMIT 20;
```

Example

```
RAM 95%
↓
mysqld consuming most RAM
↓
Buffer pool = reasonable?
↓
Connections = 4000?
↓
```

Per-session buffers?

↓

Large temporary tables?

↓

Find actual consumer

Don't immediately reduce the buffer pool. Find out where the memory is actually going.

3. High Disk I/O — what can cause it ?

This is very important for MySQL.

Common causes:

Heavy SELECT → table scans

Large INSERT/UPDATE/DELETE

Large purge/delete job

InnoDB flushing dirty pages

Redo-log pressure

Undo/purge activity

Temporary tables going to disk

Backup running

Replication activity

Slow storage / insufficient IOPS

Insufficient buffer pool causing more disk reads

What I check

Linux:

iostat -xz 1

Pay attention to:

%util

await

r/s

w/s

Also:

iotop

if available.

Then:

vmstat 1

And:

df -h

to make sure you aren't running out of space.

4. If MySQL is doing high I/O

Check:

SHOW ENGINE INNODB STATUS\G

Also:

SHOW GLOBAL STATUS LIKE 'Innodb_buffer_pool%';

Check temporary tables:

SHOW GLOBAL STATUS LIKE 'Created_tmp%';

Check running queries:

SHOW FULL PROCESSLIST;

Check transactions:

SELECT

 trx_id,

 trx_mysql_thread_id,

 trx_started,

 trx_rows_modified,

 trx_rows_locked,

 trx_query

```
FROM information_schema.innodb_trx
ORDER BY trx_started;
```

A huge transaction can be a major source of I/O.

5. High I/O during your data purge scenario

This connects directly to what we discussed earlier.

Suppose someone runs:

```
DELETE FROM DatabaseConnectionLog
WHERE created_at < NOW() - INTERVAL 90 DAY;
```

and it deletes 500 million rows.

You could get:

Huge DELETE



UNDO generated



REDO generated



Dirty pages



InnoDB flushing



Heavy disk I/O



Replication lag



Production slowdown

That's why you'd use small batches and monitor the server.

6. What if CPU + Memory + I/O are ALL high ?

This is where interviewers really test you.

Don't treat them as three separate problems.

For example:

CPU 95%

Memory 90%

Disk I/O 95%

Could be caused by one bad query.

Example:

Bad query



Full table scan



Reads huge amount of data from disk



CPU processes millions of rows



Sort/join requires memory

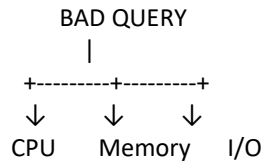


Temporary table



Disk I/O

So:



This is why you need to find the root cause, rather than independently tuning CPU, memory and disk.

7. My production checklist

If an alert says:

"MySQL server performance is degraded."

I'd follow this:

Step 1 — OS

uptime

free -h

top

vmstat 1

iostat -xz 1

df -h

Step 2 — Find whether MySQL is responsible

```
ps -eo pid,user,%cpu,%mem,rss,cmd --sort=-%cpu | head
```

Step 3 — Check MySQL workload

```
SHOW FULL PROCESSLIST;
```

Step 4 — Check transactions

```
SELECT * FROM information_schema.innodb_trx;
```

Step 5 — Check locks

```
SELECT *
```

```
FROM performance_schema.data_lock_waits;
```

Step 6 — Check InnoDB

```
SHOW ENGINE INNODB STATUS\G;
```

Step 7 — Find expensive SQL

Use Performance Schema / sys schema and then:

```
EXPLAIN <query>;
```

Step 8 — Take corrective action

Depending on what you found:

Kill runaway query

OR

Fix/optimize query

OR

Add appropriate index

OR

Reduce excessive connections

OR

Throttle batch DELETE

OR

Stop/ reschedule backup

OR

Address disk/storage problem

OR

Tune MySQL configuration

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